

Climate Finance

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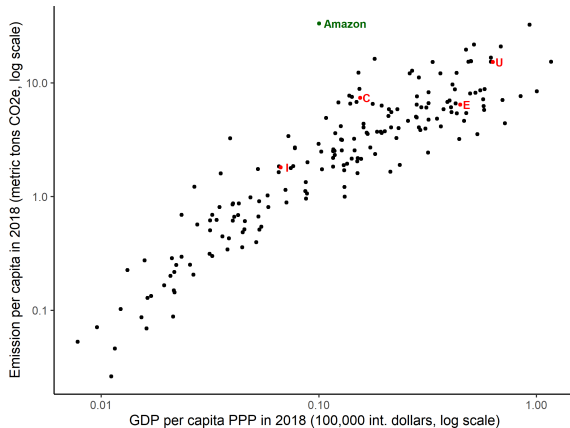
Columbia University

PhD course, Spring 2025

Motivation

- The Amazon forest contains 123 ± 23 billion tons of captured carbon that can be released to the atmosphere.(above+below ground).
- Brazilian Amazon occupies 60% of the 2.7 million square miles that comprise the Amazon.
- An area the size of Texas has been deforested in the Brazilian Amazon.
- Portions of Amazon have become a source instead of sink for carbon.

Emissions curve



This research

- Assunção et al. [2025].
- Studies the problem of a **fictitious social planner** to provide a benchmark for *ad hoc* policy alternatives.
- Analyzes a dynamic model across heterogeneous regions in the Amazon.
- Exploits rich panel data set that covers a cross-section of regions in the Amazon.
- Uses numerical methods to achieve a necessary degree of economic and environmental richness to achieve credible results.
- Implements a novel refinement to uncertainty quantification.

Plan

- ① Present basic model
- ② Discuss calibration
- ③ Present results
- ④ Parameter ambiguity
 - Computation using Hamiltonian Monte Carlo
- ⑤ Interactions across sites

State and control variables

- Sites $i = 1, \dots, I$ and state-vector (Z, X, P^a) .
- $Z = (Z^1, \dots, Z^I)$, is the vector of site-specific hectares of land used for agriculture.
- $X = (X^1, \dots, X^I)$ is the vector of site-specific stocks of captured carbon (above ground).
- P^a is an index of cattle prices in Brazil in 2017 USD.
 - 85% of deforested land is used for cattle raising.
- P^e is the social price of emissions
- Non-negative controls U^i and V^i with $\dot{Z}_t = U_t - V_t$
 - At optima $U_t^i V_t^i = 0$
- State constraints:

$$0 \leq Z^i \leq \bar{z}^i,$$

where $\bar{z}^i$ is the maximum area for agriculture in site i .

State dynamics

- Carbon capture dynamics

$$\dot{X}_t^i = -\gamma^i U_t^i - \alpha X_t^i + \alpha \gamma^i (\bar{z}^i - Z_t^i)$$

where $\gamma^i > 0$ and $\alpha > 0$.

- Does not allow for interactions across sites.
- q state Markov chain with possible values for the agricultural price, $p_1^a, \dots, p_q^a$
- An infinitesimal generator given by $q \times q$ matrix $\mathbb{M} = [m_{\ell, \ell'}]$ with non-negative off-diagonal entries and

$$\sum_{\ell' \neq \ell} m_{\ell \ell'} = -m_{\ell \ell} > 0.$$

Outputs

- Agricultural output

$$A^i = \theta^i Z^i$$

where $\theta^i \geq 0$ is a site specific productivity parameter.

- Net emissions

$$\kappa \sum_{i=1}^I Z^i - \sum_{i=1}^I \dot{X}^i$$

where $\kappa > 0$ measures the emissions per hectare of land induced by agriculture.

Land-use change cost

- Quadratic adjustment costs

$$LC = \frac{\zeta_1}{2} \left(\sum_{i=1}^I U^i \right)^2 + \frac{\zeta_2}{2} \left(\sum_{i=1}^I V^i \right)^2.$$

Social planner's objective

- Planner maximizes

$$\mathbb{E} \int_0^{\infty} e^{-\delta t} \left[P_t^a \sum_{i=1}^I A_t^i - P^e \left(\sum_{i=1}^I \kappa Z_t^i - \dot{X}_t^i \right) - L C_t \right] dt$$

- Planner chooses site-specific controls U^i, V^i subject to the state evolution equations and the initial states
- P^e , price of emissions, reflects a market for offsets and/or a planner's own valuation.
- Cross sectional heterogeneity often implies boundary solutions for sites.

Adding parameter uncertainty

- $P_t^a \equiv P^a$, $\varphi^i := (\gamma^i, \theta^i)$, φ full $2I$ dim parameter vector.
- $\varphi = \varphi(\rho)$, $\dim(\rho) < 2I$.
- π baseline distribution of ρ .
- d be the vector of decisions and $f(d, \varphi(\rho))$ for the resulting value given the unknown ρ .
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$$\max_d \min_{g, \int g d\pi = 1} \int f(d, \rho) g(\rho) d\pi(\rho) + \xi \int \log g(\rho) g(\rho) d\pi(\rho)$$

- $\xi > 0$ is penalty parameter, and minimizing g is given by:

$$g_d(\rho) = \frac{\exp\left[-\frac{1}{\xi} f(d, \rho)\right]}{\int_{\mathcal{B}} \exp\left[-\frac{1}{\xi} f(d, \rho)\right] d\pi(\rho)} \quad (1)$$

Deterministic limit case

- Limit case with $\tilde{\zeta} = \infty$ corresponds to *ambiguity neutrality*. Decision problem uses objective

$$\max_d \int_{\mathcal{B}} f(d, \rho) g(\rho) d\pi(\rho).$$

- Problem is deterministic with parameters set at the average value of ρ under the baseline distribution π .

Sites and initial states

- Sites:
 - Grid of 1887 sites of $\approx 67 \text{ km} \times 67 \text{ km}$. 1043 $\geq 3\%$ area in the Brazilian biome. Results without price uncertainty
- Agricultural areas in 2017 (Z_0^i)
 - Source: MapBiomas
- Total land available in 2017 ($\bar{z}^i$)
 - Source: MapBiomas
- $X_0^i = \gamma^i(\bar{z}^i - Z_0^i)$.

Agriculture productivity

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- Value of cattle sales and area of agriculture for 558 municipalities that overlap the biome from 2017 Agriculture Census.
- Missing data and unlikely outliers for municipalities with very small agricultural area (92 out of 558 observations).
- Regression of (log) value of cattle/hectare on geographical variables R_{θ}^m with random effects at a coarser spatial level yields smoother and lower dimensional representation and fills in missing data.

Agriculture productivity



- θ^i : weighted (by agricultural area) average of predicted output per acre over overlapping municipalities, divided by P_{2017}^A .
- Heterogeneity reflects transportation cost and suitability of site.
- Agricultural price uncertainty: Fit two-state Markov process as hidden state Markov chain with Gaussian noise, using **hmmlearn** package in python. P^S mean of stationary distribution.

Agriculture productivity



- Land-use change parameter ζ_1 , so marginal cost of changing land use matches average difference in prices between forested and naked land.
- Land-use change ζ_2 , so marginal cost of reforestation matches survey of cost of natural reforestation.

Carbon dynamics

- γ^i : Extract random sample of 1.2M 30m-pixels and select 893,753 pixels that could be considered **primary forest** in 2018 (pixels with no deforestation at least since 1985). Add above ground biomass density data for 2017, from ESA Biomass (Santoro and Cartus [2021]). Biomass data comes in a grid format $\sim 100\text{m}$, so spatially match it to sample and calculate average CO_2 density (Mg/ha).
- Calculate mean γ^i for each fine site $i = 1, \dots, I$.
- Regression on geographical variables (with random effects at the coarse site level) yields smoother representation and fills in missing data.

Carbon dynamics

II

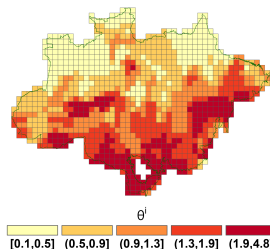
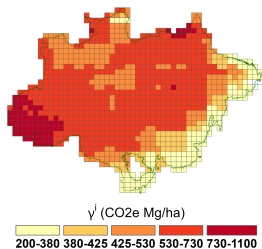
- α , carbon depreciation parameter, set so convergence of carbon accumulation process is 100 years. (Henrich et al.(2021)).
- κ calibrated from agricultural net annual emission data at the state level available from SEEG.

Baseline distribution

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- Use Gibbs sampling with diffuse priors (Hansen and Sargent [2013], Section 5.3) to produce a baseline distribution π of the vector β , which is used for the case of parameter uncertainty.

Site-specific Parameters γ^i and θ^i (1043 sites)



Computational Approach

- Discrete-time (year) approximation
- Deterministic prices: Interior Point Method: inequalities are approximated with logarithmic penalty functions.
- Uncertain prices: Add Model Predictive Control:
 - Finite-horizon approximation with two horizons:
 - Relatively short uncertainty horizon (u.h.) where controls are computed as a function of potential shock realizations (five periods);
 - Longer horizon where the control solutions are approximated by eliminating shocks beyond the uncertainty horizon (200 periods).
 - Solve the model again in subsequent periods with the same u.h..

Computational Approach

II

- Choose $u.h. = 5$ because value function changes little from $u.h. = 4$.
- Interested in first 50 years

Parameter uncertainty

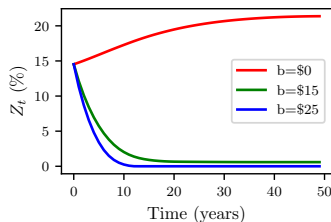
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- i) Given a g , solve the maximization problem for a candidate d . May ignore relative entropy penalty.
- ii) For given d , solve the minimization problem to obtain new g .
- iii) Repeat until achieve convergence.
 - For step ii) use quasi-analytical solution (1) and Markov chain Monte Carlo method that is based on Hamiltonian dynamics. Often more efficient for high dimensional problems than Metropolis-Hastings (Neal et al. [2011], Carpenter et al. [2017]).

Planner's own valuation - Inferred shadow price of carbon from aggregate behavior 1995-2008

- 1995 - Begin reliable price data. 2008 - Announcement of Amazon Fund that incentivizes preservation of the forest with resources from Norway. (NOK 8.3 billion in 2009-2018)
- Observe prices P_t^a in 95-08 and find P^{ee} that produces $Z_{2008} = Z_{2008}^o$.
- Price P^{ee} that matches observed deforestation varies with model chosen.
- A model where, implicitly, a planner would act more aggressively against preservation would imply a larger P^{ee} .
- Larger P^{ee} applied to future decisions lowers deforestation (increase reforestation).

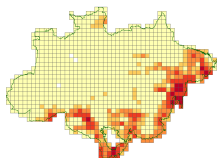
Evolution of agricultural area (deterministic case, $P^a = P^s$)



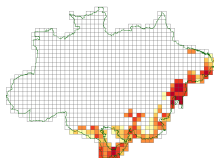
1043 Sites

- $P^{ee} = \$6.6$.
- Business as usual agr. area $\sim 25\%$ - may result on tipping of east, south and central Amazon (Lovejoy and Nobre [2018]).

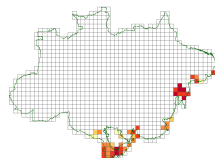
Evolution of occupation by agriculture, 1043 sites, $b = 15, P^a = P^s$



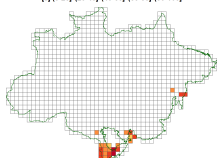
Z_{2017}^{i} (%)
[0] [0-20] [20-40] [40-60] [60-80] [80-100]



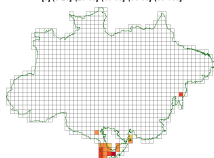
Z_{2022}^{i} (%), $b=15$
[0] [0-20] [20-40] [40-60] [60-80] [80-100]



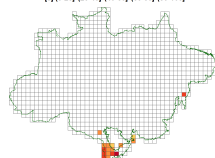
Z_{2027}^{i} (%), $b=15$
[0] [0-20] [20-40] [40-60] [60-80] [80-100]



Z_{2032}^{i} (%), $b=15$
[0] [0-20] [20-40] [40-60] [60-80] [80-100]



Z_{2037}^{i} (%), $b=15$
[0] [0-20] [20-40] [40-60] [60-80] [80-100]



Z_{2047}^{i} (%), $b=15$
[0] [0-20] [20-40] [40-60] [60-80] [80-100]

Planner Value Decomposition (200 years)

Table: 1043 Sites - Deterministic case

P^e (\$)	b (\$)	agricultural output value (\$ billion)	net transfers (\$ billion)	forest services (\$ billion)	adjustment costs (\$ billion)	planner value (\$ billion)
6.6	0	360	0	-113	6	242
16.6	10	73	135	89	10	288
21.6	15	37	235	104	15	361
26.6	20	19	332	109	18	442
31.6	25	15	422	111	22	526

Notes: Present-value decomposition under ambiguity neutrality. We set $P^a = \$41.1$, which is the mean agricultural price in the stationary distribution. Forest services are calculated using baseline shadow price ($b = \$0$). The present values are computed for two hundred years.

Transfer cost (30 years, 1043 sites)

P^e (\$)	b (\$)	15 years			30 years		
		net captured emissions (CO ₂ e Gt)	discounted net transfers (\$ billion)	effective cost (\$ per ton of CO ₂ e)	net captured emissions (CO ₂ e Gt)	discounted net transfers (\$ billion)	effective cost (\$ per ton of CO ₂ e)
6.6	0	-11.6	0	-	-15.9	0	-
16.6	10	7.1	59	3.8	14.4	108	4.8
21.6	15	9.1	116	6.6	16.7	191	7.7
26.6	20	9.9	169	9.2	17.6	271	10.5
31.6	25	10.5	224	11.8	18.0	348	13.3

Notes: Transfer costs under ambiguity neutrality. We set $P^a = \$41.1$, which is the mean agricultural price in the stationary distribution.

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